

Financial Fraud Detection Dashboard – Project Report

Project Title

Financial Fraud Detection Dashboard using Power BI

1. Project Overview

The Financial Fraud Detection Dashboard is an interactive Power BI solution developed to monitor financial transactions, identify fraudulent activities, and provide insights into fraud trends across different states and merchant categories.

The dashboard enables business users, fraud analysts, and management teams to quickly identify suspicious transactions, analyze fraud distribution, and make informed decisions to reduce financial losses.

2. Objective

- Monitor total financial transactions.
- Identify fraudulent transactions.
- Calculate fraud percentage.
- Track fraud amount.
- Analyze transaction amount by state.
- Compare fraud across merchant categories.
- Monitor monthly transaction trends.
- Enable interactive filtering using slicers.

3. Dataset Information

Dataset Name: Cleaned Fraud Dataset

Total Records: 5,000

Total Columns: 23

Domain: Banking / Financial Fraud Detection

Tool Used: Power BI

Important Fields:

Transaction ID, Transaction Date & Time, Merchant, Merchant Category, Transaction Amount, Fraud Status, Fraud Label, State, City, Gender, Job, Age, Month, Weekday, Hour

4. Dashboard KPIs

Total Transactions: 5,000

Total Transaction Amount: 6.88M

Total Fraud Transactions: 500

Fraud Percentage: 10%

Fraud Amount: 1.23M

5. Dashboard Visualizations

1. KPI Cards

* The KPI cards provide a quick summary of the overall financial performance by displaying key metrics such as *Total Transactions, Total Transaction Amount, Total Fraud Transactions, Fraud Percentage, and Fraud Amount*. These indicators help users understand the current fraud situation at a glance.

* They enable management to monitor business performance in real time, identify increases in fraudulent activities, and make faster, data-driven decisions without analyzing detailed reports.

2. Map Visualization – Total Transaction Amount by State

* The map visual displays the total transaction amount across different states, making it easy to compare regional transaction volumes and identify locations with higher financial activity.

* It helps fraud analysts detect geographical patterns, monitor high-value transaction regions, and prioritize fraud investigation efforts in states with significant transaction amounts.

3. Line Chart – Total Transaction Amount by Month

* The line chart illustrates the monthly trend of total transaction amounts, allowing users to analyze how transaction values change throughout the year.

* It helps identify seasonal trends, unusual spikes or declines in transaction activity, and supports forecasting and financial planning.

4. Column Chart – Fraud Amount by Merchant Category

* This chart compares the total fraud amount across different merchant categories such as *Electronics, Grocery, Shopping, Travel, Gas, and Entertainment*, helping identify categories with the highest fraud losses.

* It enables businesses to recognize high-risk merchant categories and implement stronger fraud detection and preventive measures where they are needed most.

5. Stacked Column Chart – Total Transactions by Merchant Category and Fraud Label

* The stacked column chart compares the number of *Fraud* and *Genuine* transactions within each merchant category, providing a clear view of transaction distribution.

* It helps evaluate fraud occurrence across different business sectors, identify categories with relatively higher fraud rates, and measure the effectiveness of fraud prevention strategies

6. Donut Chart – Total Transaction Amount by Merchant Category

* The donut chart displays the percentage contribution of each merchant category to the overall transaction amount, making it easy to understand the distribution of transaction values.

* It helps management identify the merchant categories contributing the highest transaction volume and assess whether high transaction volume is associated with increased fraud risk.

6. Dashboard Filters (Slicers)

State

Merchant Category

Fraud Label

7. Key Insights

- 5,000 transactions analysed.
- Total transaction value: 6.88M.
- 500 fraudulent transactions.
- Fraud rate: 10%.
- Fraud amount: 1.23M.
- Shopping and Travel are among the higher fraud-value categories.
- Monthly transactions remain relatively stable.
- Genuine transactions significantly outnumber fraudulent transactions.

8. Business Benefits

Detect fraud quickly, reduce financial losses, monitor fraud trends, identify high-risk merchant categories, analyse fraud geographically, support investigations, and improve fraud prevention.

9. Tools & Technologies Used

Power BI Desktop

Power Query

DAX

Data Modelling

Interactive Visualisations

Slicers and Filters

10. Skills Demonstrated

Data Cleaning

Data Transformation

Data Modelling

DAX

KPI Development

Dashboard Design

Business Intelligence Reporting
Data Visualisation
Financial Data Analysis
Fraud Analytics

11. Conclusion

The Financial Fraud Detection Dashboard provides a comprehensive overview of transaction activity and fraud performance. It helps stakeholders identify fraud patterns, monitor business performance, and make informed decisions to strengthen fraud detection and minimise financial risk.